

Bayesian Hill Mixture Models for Heterogeneous Consumer Response in Marketing Mix Modeling

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Abstract

Marketing Mix Models (MMM) are widely used for measuring advertising effectiveness, yet standard implementations assume homogeneous consumer response to marketing spend. This paper proposes a Bayesian mixture of Hill saturation functions to capture heterogeneous response patterns across latent consumer segments—a capability that single-curve models do not provide. We implement the approach in NumPyro with automatic prior scaling and label-invariant convergence diagnostics with post-hoc relabeling. Our confirmatory evidence comes from a calibrated synthetic benchmark of 80 runs across four data-generating processes. When the true response is homogeneous, the single-Hill baseline remains best and mixture models incur only a small ELPD penalty (about 1 point). When heterogeneity is present, the matched mixture improves ELPD by roughly +3, +10, and +18 for true $K = 2, 3, 5$ respectively. We evaluate holdout predictions sequentially, preserving both adstock state and absolute time index across the train/test split, and find that mixture gains are strongest in predictive density while RMSE differences remain smaller and latent-state coverage becomes difficult in the highest- K settings. Real-data experiments are currently treated as exploratory because data selection and preprocessing are still under refinement. We provide practical guidance for using mixture-based MMM as a controlled benchmark for heterogeneous saturation effects.

Keywords: Marketing Mix Modeling, Bayesian inference, Hill function, mixture models, label switching

1 Introduction

Marketing Mix Modeling (MMM) enables organizations to quantify the effectiveness of marketing investments and optimize budget allocation. Modern implementations typically employ Hill saturation functions to capture diminishing returns and geometric decay adstock transformations to model carryover effects (Jin et al., 2017; Chan and Perry, 2017). We refer to this standard approach—using a single Hill curve to model the relationship between marketing spend and response—as *Standard MMM*.

A critical limitation of Standard MMM is its assumption that all consumers respond identically to marketing stimuli. In reality, different segments—heavy versus light buyers, brand loyalists versus switchers—exhibit heterogeneous response patterns (Wedel and Kamakura, 2000; Allenby and Rossi, 1998). A single aggregate response curve represents a weighted average that masks segment-specific behaviors, potentially leading to suboptimal budget allocation. Recent work has shown that apparent nonlinear effects in MMM may be artifacts of this homogeneity assumption (Dew et al., 2024), further motivating more flexible approaches.

This paper proposes *Mixture MMM*: a Bayesian mixture of Hill saturation functions that captures heterogeneous consumer response across latent segments. The key insight is that mixture

models—regardless of the specific number of components—provide a principled framework for modeling response heterogeneity that Standard MMM cannot capture. While the optimal number of mixture components K depends on dataset characteristics, the fundamental advantage of the mixture approach is consistent: the ability to identify and quantify distinct response patterns within the consumer population.

We implement Mixture MMM using NumPyro (Phan et al., 2019) with JAX acceleration, which recent benchmarks show achieves 2–20x faster sampling than TensorFlow-based alternatives while maintaining superior channel contribution recovery (PyMC Labs, 2025). A key methodological contribution is our treatment of label switching in mixture posteriors. Standard convergence diagnostics (R-hat) are unreliable for mixture models because component labels may exchange across MCMC chains. We develop post-hoc relabeling and label-invariant diagnostics to make mixture benchmarks interpretable. In this version of the paper, we ground our empirical claims primarily in synthetic experiments with known truth; real-data checks are reported only as exploratory evidence while data curation and preprocessing are still being standardized.

2 Model Specification

Let x_t denote marketing spend at time t and y_t the observed outcome. The model proceeds as follows.

Adstock transformation. We apply geometric decay to capture carryover effects:

$$s_t = x_t + \alpha \cdot s_{t-1}, \quad s_0 = 0 \quad (1)$$

where $\alpha \in [0, 1]$ is the decay parameter with prior $\alpha \sim \text{Beta}(2, 2)$.

Hill saturation. For each latent segment $k \in \{1, \dots, K\}$, the response function is:

$$f_k(s) = A_k \cdot \frac{s^{n_k}}{\lambda_k^{n_k} + s^{n_k}} \quad (2)$$

where A_k is the maximum effect, λ_k is the half-saturation point, and $n_k \sim \text{LogNormal}(\log 1.5, 0.4)$ controls curve steepness.

Mixture likelihood. The observation model is a Gaussian mixture:

$$y_t \sim \sum_{k=1}^K \pi_k \cdot \mathcal{N}(\mu_0 + \beta t + f_k(s_t), \sigma^2) \quad (3)$$

where $\pi \sim \text{Dirichlet}(\mathbf{1}_K)$ and μ_0, β capture baseline trend.

Automatic prior scaling. To accommodate diverse data scales, priors are computed automatically from training data: $A_k \sim \text{LogNormal}(\log(0.3 \cdot \text{range}(y)), 0.8)$, $\mu_0 \sim \mathcal{N}(\bar{y}, 2\sigma_y)$, and $\sigma \sim \text{HalfNormal}(\sigma_y)$.

2.1 Identifiability and Label Switching

Mixture models suffer from label switching, where posterior samples may exchange component labels across MCMC iterations. This renders standard convergence diagnostics meaningless: chains may have converged to the same posterior but with permuted labels.

Within-MCMC ordering. We impose ordering via cumulative sum reparameterization: $\lambda_k = \sum_{j=1}^k \delta_j$ with $\delta_j \sim \text{LogNormal}(\log(s_{\max}/(K+1)), 0.7)$, ensuring $\lambda_1 < \lambda_2 < \dots < \lambda_K$. However, this approach may distort the posterior geometry.

Post-hoc relabeling. We implement unconstrained sampling followed by post-hoc relabeling: after MCMC, samples are permuted so that $\lambda_1 < \lambda_2 < \dots < \lambda_K$ at each iteration. This preserves detailed balance while enabling meaningful diagnostics.

Label-invariant diagnostics. We compute R-hat on three quantities: (1) the total log-likelihood (label-invariant by construction), (2) scalar parameters (α , σ), and (3) relabeled component parameters. A model is considered converged when all three criteria yield R-hat < 1.01 .

3 Experiments

We evaluate the proposed models primarily through synthetic experiments with known ground truth. Real-data checks are retained only as exploratory diagnostics in the current draft.

3.1 Synthetic Data Experiments

Data generation. We evaluate model performance across four data-generating processes (DGPs) with varying complexity: (1) **Single** ($K = 1$): standard single Hill response, (2) **Mixture** $K = 2$: two-component mixture, (3) **Mixture** $K = 3$: three-component mixture, and (4) **Mixture** $K = 5$: five-component mixture. Each DGP uses $T = 200$ observations with a 75/25 train/test split. We run 5 random seeds per condition (80 total experiments).

Model comparison. We compare four specifications: (1) **Single Hill** with one global saturation curve, (2) **Mixture** $K = 2$ with two components, (3) **Mixture** $K = 3$ with three components, and (4) **Mixture** $K = 5$ with five components. Inference uses NUTS with 1,000 warmup and 2,000 sampling iterations across 4 chains.

Evaluation protocol. Holdout predictions are generated sequentially: the test period inherits the adstock state accumulated during training and is evaluated at its absolute time index rather than being re-indexed to start at zero. This avoids an artificial reset of carryover and trend at the train/test boundary. We report ELPD-LOO as the primary predictive metric, with test RMSE and posterior latent-mean recovery as secondary checks.

Results. Seventy-nine of 80 synthetic runs satisfied the effective convergence criterion. The only miss was a Mixture $K = 2$ fit under the true $K = 3$ DGP, which remained above threshold after repeated retries. For single-Hill fits we require standard R-hat < 1.05 ; for mixture fits we require both label-invariant and relabeled diagnostics to satisfy the stricter $\hat{R} < 1.01$ threshold.

Table 1 shows predictive performance across the representative model comparisons.

When the true DGP is simple ($K = 1$), mixture models incur only a marginal ELPD penalty. For heterogeneous data ($K \geq 2$), the mixture family becomes decisively better, and the gain increases with true complexity. The strongest improvements appear in ELPD rather than RMSE, which suggests that mixture models primarily improve probabilistic calibration rather than only point prediction.

Table 1: Predictive performance on synthetic data (mean $\pm$ standard deviation across 5 seeds).

True K	Model	ELPD-LOO	Test RMSE	Δ LOO vs Single
$K = 1$	Single Hill	-382.4 ± 11.9	3.0 ± 0.4	—
$K = 1$	Mixture $K = 2$	-383.4 ± 12.0	3.0 ± 0.4	-1.1
$K = 2$	Single Hill	-386.7 ± 7.5	3.1 ± 0.4	—
$K = 2$	Mixture $K = 2$	-383.6 ± 9.5	3.2 ± 0.4	+3.2
$K = 2$	Mixture $K = 3$	-383.8 ± 9.3	3.2 ± 0.4	+2.9
$K = 3$	Single Hill	-463.1 ± 7.8	4.9 ± 0.4	—
$K = 3$	Mixture $K = 3$	-452.7 ± 8.7	5.0 ± 0.4	+10.5
$K = 3$	Mixture $K = 5$	-453.1 ± 8.7	5.0 ± 0.4	+10.0
$K = 5$	Single Hill	-463.8 ± 8.4	4.8 ± 0.4	—
$K = 5$	Mixture $K = 3$	-445.6 ± 5.9	4.8 ± 0.3	+18.1
$K = 5$	Mixture $K = 5$	-445.6 ± 5.4	4.8 ± 0.3	+18.2

Latent-mean recovery adds an important nuance. For the simpler DGPs ($K = 1, 2$), the matched mixture achieves test μ RMSE below 1.0 with 90% coverage between 0.81 and 1.00. For the harder $K = 3$ and $K = 5$ DGPs, the matched mixture still wins on ELPD, but test μ RMSE remains around 4.1 and 90% coverage drops to roughly 0.22–0.25. In other words, predictive density improves before latent-state uncertainty becomes well calibrated in the highest-heterogeneity settings.

The synthetic benchmark also exposes a useful model-selection pattern. The mixture model with the correct K tends to win on ELPD, but modest over-specification remains competitive because excess components are only partially used. This is encouraging for practice: slight over-specification appears less harmful than enforcing a single saturation curve when the truth is heterogeneous.

3.2 Exploratory Real-Data Check

We retain a preliminary real-data benchmark on the Conjura MMM dataset only as an exploratory check. At the current stage, organization selection, preprocessing, and covariate construction are still under refinement, and many runs exhibit poor PSIS-LOO diagnostics and strong seed sensitivity. We therefore do not use the real-data results as confirmatory evidence in this paper. Their present role is limited to stress-testing the implementation and identifying the next methodological bottlenecks for future work.

4 Discussion

The synthetic benchmark establishes a clean and asymmetric result: the single-Hill model is preferred when the data are truly homogeneous, but the penalty for fitting a small mixture is minor; when heterogeneity is present, the single-curve restriction becomes clearly suboptimal and mixture models gain predictive density rapidly as true complexity increases.

Standard MMM vs Mixture Models. The central finding is therefore not that one fixed K dominates everywhere, but that the mixture family is the correct modeling language once heterogeneous saturation is plausible. In the current benchmark, the correct- K mixture typically wins on ELPD, while simpler or slightly over-specified mixtures often remain close. Standard MMM re-

mains a strong baseline for homogeneous settings, but it cannot represent multi-segment response curves.

Component selection trade-offs. The choice of K is best understood as a trade-off between expressiveness and simplicity. Larger K helps when the DGP is genuinely heterogeneous, especially in predictive density, but the incremental benefit in RMSE is smaller. This suggests that additional components mostly improve uncertainty quantification and distributional fit rather than only mean prediction. At the same time, latent-state recovery is a stricter target than predictive density: in the $K \geq 3$ DGPs, matched mixtures still dominate ELPD, but 90% test- μ coverage remains far below nominal.

Relabeling as a diagnostic tool. Post-hoc relabeling remains important even in a synthetic suite with strong convergence. Its main value is not to manufacture convergence, but to keep diagnostics interpretable and to separate genuine non-identifiability from label permutations. For mixture MMM, reporting only parameter-level R-hat without a label-invariant check is not sufficient.

When to use Standard vs Mixture MMM. The choice between standard and mixture models depends on the application context and data characteristics:

- **Standard MMM** is appropriate when (a) consumer response is plausibly homogeneous, (b) a single interpretable curve is preferred, or (c) the application does not justify mixture complexity.
- **Mixture MMM** is appropriate when (a) heterogeneous saturation is substantively plausible, (b) predictive density and uncertainty quantification matter, or (c) segment-specific response curves are part of the scientific question.

Our experiments suggest that small mixtures incur only modest penalty when heterogeneity is absent, but practitioners should still weigh that robustness against the simpler interpretation of a single-curve model.

Practical recommendations for mixture models.

1. Start with Single Hill and Mixture $K = 2$ as the minimum comparison set.
2. Escalate to larger K only when predictive-density gains justify the additional complexity.
3. Always evaluate holdout predictions sequentially, preserving adstock state and absolute time index across the split.
4. Always report label-invariant and relabeled diagnostics for mixture models.

Limitations. Our approach assumes segment membership is constant over time, which may not hold during product launches or competitive shifts. The current synthetic suite is deliberately in-model and single-channel, so it does not yet probe robustness to misspecification, multiple media channels, or aggressive prior mismatch. The mixture structure also increases computational cost compared to single-curve models. Finally, the real-data benchmark is still exploratory; stronger claims will require more disciplined preprocessing and broader external validation.

5 Conclusion

We presented a Bayesian mixture of Hill saturation functions as an alternative to standard single-curve Marketing Mix Models. Our central finding is that mixture models offer a principled approach to capturing heterogeneous consumer response—a capability fundamentally absent from traditional MMM implementations.

In a controlled synthetic benchmark, mixture models improve ELPD whenever the true response is heterogeneous, with gains that grow with latent complexity, while paying only a small penalty when the population is homogeneous. This makes the mixture formulation a credible benchmark for studying heterogeneous saturation effects under known ground truth.

Our methodological contribution—label-invariant diagnostics with post-hoc relabeling and sequential holdout evaluation—addresses two practical barriers to studying mixture MMMs: interpreting convergence and evaluating out-of-sample predictions without resetting latent state. These details matter because they determine whether predictive comparisons are scientifically meaningful. The updated synthetic benchmark also shows an important boundary: predictive gains do not automatically imply calibrated latent-state uncertainty in the most heterogeneous settings.

The choice of mixture complexity (K) should be guided by data characteristics and application requirements. Larger K captures richer heterogeneity, but the main benefit appears first in predictive density rather than RMSE. Rather than prescribing a single “best” K , we recommend practitioners compare a small set of models, monitor label-invariant diagnostics, and treat current real-data evidence as exploratory until data preparation is better standardized.

Implementation in NumPyro with automatic prior scaling and comprehensive diagnostics provides a practical tool for practitioners. Code and data are available at <https://github.com/jpcca/marketing-mix-modelling>.

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